

YOGABALAN B R

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OBJECTIVE

Computer Science student applying for a Software Engineering Internship. I build full-stack applications with React, TypeScript, Node.js and Java (Spring Boot), and I also work on embedded projects in C/C++ and Python. Won 1st place at a college hackathon and led a team through a 15-day internship project.

EDUCATION

KSR College of Engineering, Tiruchengode 2024 – Present
B.E. Computer Science and Engineering | CGPA: 7.98 / 10
Sowdambikaa Matric. Boys Hr. Sec. School, Thuraiyur 2024
Higher Secondary | 79%

TECHNICAL SKILLS

Programming: Java, C, C++, Python, TypeScript, JavaScript, SQL
Web & Backend: React, Node.js, Express, Spring Boot, REST APIs, JWT Authentication, TailwindCSS
Databases: PostgreSQL, Firebase (Auth, Firestore), Supabase, Prisma ORM, SQLite
Embedded & IoT: ESP32, Raspberry Pi, Arduino, ESP-IDF, MPU6050, PID Control, OpenCV
Tools: Git, GitHub, Docker, Linux, VS Code, MATLAB/Simulink
Spoken Languages: English, Tamil, Hindi, Telugu

EXPERIENCE

Team Lead Intern, TouchMark Descience Pvt Ltd Jun 2026 – Jul 2026
Remote

- Led a team in a 15-day internship to build an Inventory Management System using React, Spring Boot and PostgreSQL.
- Split the work between frontend and backend members and handled integration so the application was working by the end of the internship.

PROJECTS

Team-5 Inventory Management System | [GitHub](#) Team Project
React 18, Material UI, Recharts, Firebase Authentication, Cloud Firestore

- Built a serverless inventory web app covering sales, purchases, returns, accounts and reports. The React frontend talks directly to Firebase, so no backend server is needed.
- Set up role-based access for six roles (Admin, Store Manager, Billing Clerk, Purchase Manager, Accounts, Staff) using Firestore security rules.
- Used Firestore transactions to issue unique document numbers and update stock together, so concurrent entries do not clash. Deployed on Firebase Hosting.

QR-Based Check-In / Check-Out Attendance System | [GitHub](#)
React (TypeScript), Node.js, Express, PostgreSQL, Prisma, JWT

- Built a full-stack attendance system with a React/TypeScript frontend and a REST API in Node.js and Express.
- Added JWT login, bcrypt password hashing and Zod request validation.
- Used unique database indexes to stop duplicate records when many users check in or out at the same time.

Smart Voter Verification System | [GitHub](#) April 2026
Python, OpenCV, Raspberry Pi, SQLite, Embedded Linux

- Built an offline voter verification device on Raspberry Pi that checks an Aadhaar ID and matches the voter's face in real time.
- Created separate Admin, Registrar and Verifier roles to prevent duplicate voting and unauthorized edits to records.
- Won 1st place at the college "Ship Up" Hackathon.

Learning Management System for Hindi Tuition | [GitHub](#)
React (TypeScript), Supabase (PostgreSQL, Auth, Storage), TanStack Query

- Developed a learning platform with a React/TypeScript frontend and Supabase backend for authentication, file storage and data.
- Wrote Row Level Security policies to control which users can read or change each record, and used Supabase Edge Functions for AI-assisted features.

Self-Balancing Robot Ongoing
ESP32, MPU6050, PID Control

- Built a two-wheeled robot that balances using MPU6050 readings and a PID loop on an ESP32. Now upgrading from DC motors to NEMA17 steppers with DRV8825 drivers for finer control.